ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 2.01: Matrices, Determinants & Vectors

Name: _____

Q1. Given the vectors mathbf{u} = langle 3, -4 rangle and mathbf{v} = langle -2, 5 rangle , what is the vector 2mathbf{u} + mathbf{v} ?

[ ] A: langle 4, -3 rangle
[ ] B: langle 1, 1 rangle
[ ] C: langle 4, 1 rangle
[ ] D: langle 8, -13 rangle

Q2. Matrices A and B are defined as A = begin{bmatrix} 2 & k -1 & 3 end{bmatrix} and B = begin{bmatrix} 1 & 4 2 & k end{bmatrix} , where k is a real number. If the determinant of the matrix sum A + B is equal to 17 , what is the value of k ?
Note: The determinant of a 2 times 2 matrix begin{bmatrix} a & b c & d end{bmatrix} is defined as ad - bc .

[ ] A: 1
[ ] B: 2
[ ] C: 6
[ ] D: frac{19}{2}

Q3. For all real values of theta, the matrix product begin{bmatrix} costheta & sintheta end{bmatrix} begin{bmatrix} 6 & 2 2 & 3 end{bmatrix} begin{bmatrix} costheta sintheta end{bmatrix} evaluates to a 1 times 1 matrix [k]. What is the maximum possible value of k?

[ ] A: 6
[ ] B: 7
[ ] C: 9
[ ] D: 14

Q4. Given the vectors mathbf{u} = langle 3, -2 rangle and mathbf{v} = langle 1, 5 rangle , what is the vector 2mathbf{u} + mathbf{v} ?

[ ] A: langle 7, 1 rangle
[ ] B: langle 4, 3 rangle
[ ] C: langle 8, 6 rangle
[ ] D: langle 5, 8 rangle

Q5. For a 2 times 2 matrix begin{bmatrix} a & b c & d end{bmatrix}, the determinant is defined as ad - bc. Given the matrices A = begin{bmatrix} x & 2 -1 & 3 end{bmatrix} and B = begin{bmatrix} 1 & -4 3 & x end{bmatrix}, the determinant of the matrix A + B is equal to 19. If x > 0, what is the value of x?

[ ] A: 2
[ ] B: frac{5}{4}
[ ] C: 4
[ ] D: 6

Q6. Let mathbf{u} and mathbf{v} be vectors in the coordinate plane such that |mathbf{u}| = 5, |mathbf{v}| = 10, and the angle theta between them is obtuse 90^circ < theta < 180^circ. If the minimum value of |tmathbf{u} + mathbf{v}| over all real numbers t is equal to 6, what is the value of the dot product mathbf{u} cdot mathbf{v}?

[ ] A: -50
[ ] B: -40
[ ] C: -30
[ ] D: 40

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Q7. What is the value of the determinant of the matrix begin{bmatrix} 7 & -3 2 & 4 end{bmatrix} ?

[ ] A: 34
[ ] B: 22
[ ] C: -29
[ ] D: -34

Q8. In the standard coordinate plane, vector mathbf{u} = langle 2, -1 rangle and vector mathbf{v} = langle k, 5 rangle , where k is a real constant. If the magnitude of the vector mathbf{u} + mathbf{v} is equal to 5, which of the following is a possible value of k ?

[ ] A: -5
[ ] B: -1
[ ] C: 3
[ ] D: 7